FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – III
PAPER –2
TEST DATE: 01-05-2024

ANSWERS, HINTS & SOLUTIONS

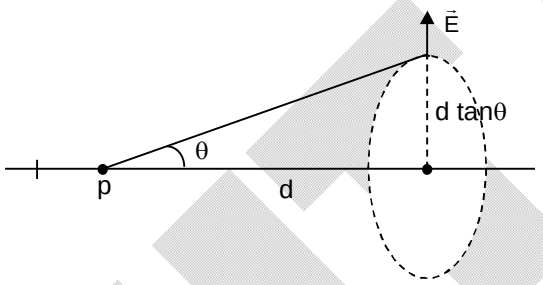
Physics

PART – I

SECTION – A

1. C

Sol. $\tan\theta = \sqrt{2}$



$$E^{net} = \frac{1}{4\pi\in_0}\frac{P}{d^3\sec^3\theta}\sqrt{1+3\cos^2\theta}$$

$$= \frac{1}{4\pi\in_0}\frac{P\sqrt{2}}{d^3\cdot 3\sqrt{3}}$$

Now $e\cdot\frac{1}{4\pi\in_0}\cdot\frac{\sqrt{2}P}{3\sqrt{3}d^3} = m\frac{v^2}{d\cdot\sqrt{2}}$

$$\Rightarrow \quad v = \sqrt{\frac{2eP}{12\sqrt{3}\pi\in_0 md^2}} = \sqrt{\frac{eP}{6\sqrt{3}\pi\in_0 md^2}}$$

2. D

Sol. Acceleration of cm of cylinder $\frac{v^2}{5R}\downarrow$

Acceleration of point P =

$$= R\cdot\left(\frac{v}{R}\right)^2 - \frac{v^2}{5R}$$

$$= \frac{4v^2}{5R}\uparrow$$

3. A

Sol. $v = \frac{K}{x} \Rightarrow F = \frac{K}{x^2}$

Conservation of mech. energy

$\frac{4K}{5R} = \frac{4K}{7R} + \frac{1}{2}mv^2$

$\Rightarrow \quad mv^2 = \frac{16K}{35R}$

Now $N - \frac{K}{(7R/4)^2} = \frac{mv^2}{R}$

$\Rightarrow \quad N = \frac{192K}{245R^2}$

4. D

Sol. $\pi R^2 \frac{H}{4} = \frac{\pi R^2 h_0}{2} \Rightarrow h_0 = \frac{H}{2}$

Now $\frac{H}{2} = \frac{\omega^2 R^2}{2g} \Rightarrow \omega \frac{\sqrt{gH}}{R}$

5. BC

Sol. Before t = 0,

Q;= CE = 4C

$i_0 \left(\text{through inductor}\right) = \frac{E}{R} = 1A$

After t = 0

$\frac{Q_i^2}{2C} + \frac{1}{2}Li_0^2 = \frac{Q_{max}^2}{2C} \Rightarrow Q_{max} = 2\sqrt{5}C$

Similarly $\frac{Q_{max}^2}{2C} = \frac{1}{2}Li_{max}^2 \Rightarrow i_{max} = \sqrt{5}A$

6. CD

Sol. B executes SHM with time period T = $2\pi\sqrt{\frac{R}{g}}$ both have same time period

When A reaches position of B, angle turned = $\frac{\pi}{3}$ At that time B will be at half of amplitude, that is $\frac{1}{2} \times \sqrt{3}\frac{R}{2}$ from O_1.

Speed $v = w\sqrt{A^2 - y^2} = \sqrt{\frac{g}{R}}\sqrt{\left(\frac{\sqrt{3}R}{2}\right)^2 - \left(\frac{\sqrt{3}R}{4}\right)^2} = \frac{3}{4}\sqrt{gR}$

When B reaches O_1 its speed is $\frac{\sqrt{3}R}{2}.\sqrt{\frac{g}{R}} = \sqrt{\frac{3gR}{2}}$

Relative speed $\sqrt{\left(\frac{\sqrt{3gR}}{2}\right)^2 + \left(\sqrt{gR}\right)^2} = \sqrt{\frac{7}{4}gR}$

7. BCD

Sol. $\frac{N_0}{5} = N_0 e^{-\lambda t_1}$

$\Rightarrow \quad t_1 = \frac{\ln 5}{\lambda}\sec$

$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{2}$

thus $t_1 = \frac{2\ln 5}{\ln 2} = 2\log_2 5\sec$

At t_2, population of N_A is 0.1 N_o

$\frac{N_o}{10} = N_o e^{-\lambda t_2}$

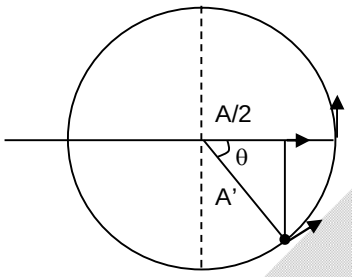
$\Rightarrow \quad t_2 = \frac{\ln 10}{\lambda} = 2\log_2 10\sec$

Now $t_2 - t_1 = 2\log_2 10 - 2\log_2 5 = 2\sec$

SECTION – B

8. 13

Sol. Speed before = $\omega\sqrt{A^2 - \frac{A^2}{4}} = \frac{\sqrt{3}A\omega}{2}$



After = $\sqrt{3}A\omega$

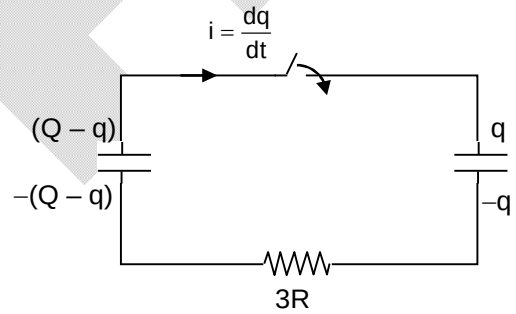
Now amplitude $A' = \frac{\sqrt{13}A}{2}$

$\cos\theta = \frac{1}{\sqrt{13}} \qquad \theta = \omega t$

$t = \frac{T}{2\pi}\cos^{-1}\left[\frac{1}{\sqrt{13}}\right]$

9. 4

Sol. $\frac{Q-q}{C} - \frac{q}{2C} - \frac{dq}{dt}\cdot 3R = 0$



$$\Rightarrow q = \frac{2Q}{3}\left(1 - e^{-\frac{t}{2RC}}\right)$$

Put $q = \frac{Q}{2}$, $t = 4RC\ell n2$

10. 9

Sol. Conservation of mech, energy

$$mg\frac{L}{2} = \frac{m}{2}g\frac{L}{4} + \frac{1}{2}\frac{mL^2}{3}\omega^2 \quad \Rightarrow \omega\sqrt{\frac{9g}{4L}}$$

11. 3

Sol. Path is cycloidal

Maximum depth = 2R

$$R = \frac{m^2g}{q^2B^2}$$

12. 5

Sol. Glass slab give constant shift.

13. 9

Sol. Critical damping

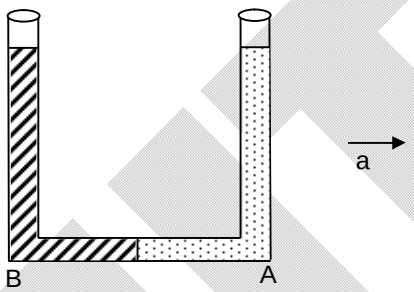
$b^2 = 4mk$

$$\Rightarrow k = \frac{b^2}{4m} = 0.09N/m$$

SECTION – C

14. 10.00

Sol. $P_BA - P_A \cdot A = \text{mass}(AB)\cdot a$



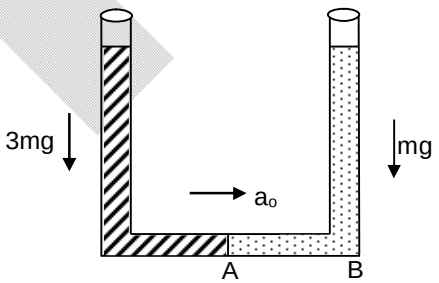
$$\Rightarrow \quad [L3\rho g - L\rho g]A = \left[3\rho\frac{L}{2}A + \rho\frac{L}{2}A\right]a$$

$$\Rightarrow \quad 2\rho Lg = 2\rho La$$

$$\Rightarrow \quad a = g = 10m/s^2$$

15. 150.00

Sol. Just after system is stopped acceleration of liquid of tube = a_o.



$m = \rho AL$

$$a_0 = \frac{2mg}{6m} = \frac{g}{3}$$

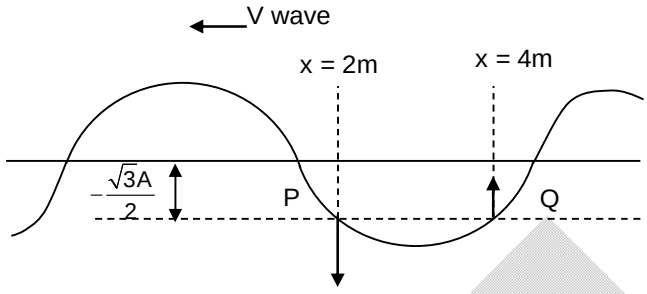
$$P_A \cdot A - mg = \frac{3m}{2} \cdot a_0$$

$$\Rightarrow P_A \cdot A = mg + \frac{3m}{2} \cdot \frac{g}{3} = \frac{3mg}{2}$$

$$\Rightarrow P_A = \frac{3}{2}\rho Lg = \frac{3}{2} \times 10^2 \times 10^{-1} \times 10 = 150 N/m^2$$

16. 12.00

Sol. $\frac{\Delta\phi}{PQ} = \frac{\pi}{3}$

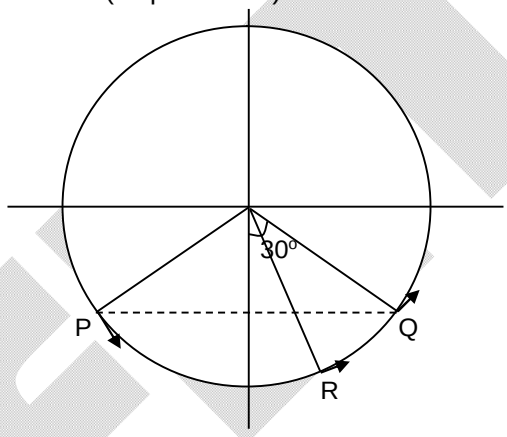


$$\Rightarrow \quad \frac{2\pi}{\lambda} \cdot 2 = \frac{\pi}{3}$$

$$\Rightarrow \quad \lambda = 12m$$

17. 49.00

Sol. x = 3.5 m (let point is 'R')



$$\Delta\phi_{RQ} = \frac{2\pi}{12} \times \frac{1}{2} = \frac{\pi}{12} rod$$

Maximum power transfer at mean position. Time to cross mean position for R is

$$t = 2 + \frac{\left(\frac{\pi}{12} + \frac{\pi}{3}\right)}{10\pi} = \frac{49}{24} \text{ second.}$$

Chemistry

PART – II

SECTION – A

18. B

Sol. $BrO_3^- + 4H^+ + 4e \longrightarrow BrO^- + 2H_2O \qquad E^0 = 0.54\ V$

$BrO^- + 2H^+ + e \longrightarrow \frac{1}{2}Br_2 + H_2O \qquad E^o = 0.45\ V$

Then E^o for $BrO_3^- + 6H^+ + 5e \longrightarrow \frac{1}{2}Br_2 + 3H_2O$

is $-5FE^0 = -4F(0.54) - F(0.45)$

$E^0 = 0.52V$

19. B

Sol. $nA_{(g)} \longrightarrow B_{(g)}$ from graph $n = \frac{1}{2}$

Here $\frac{\Delta\rho_A}{\text{time}} = \text{constant}$

It is zero order reaction

20. C

Sol. $\because \qquad \sqrt{v} = a(z-b)$

$\sqrt{v} = az - ab, \qquad ab = 1,\ a = \tan 45^o$

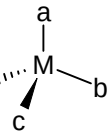
$\sqrt{v} = 51 - 1, \qquad v = 2500\ Hz$

21. D

Sol. (Reimer – Tiemann's reaction)

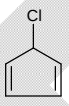
22. ACD

Sol. cis form of $\left[PtCl_2(NH_3)_4\right]^{2+}$ have super imposable mirror image B is true

Square planar have super imposable mirror image but 

is optically active.

23. BCD

Sol. (A) 

will not release Cl^- (B) $CH_3 - C \equiv CAg$ is white ppt

(C) gives CHI_3 (yellow ppt.)

(D) ppt of NaCl formed

24. ABD

Sol. Due to tautomerization.

SECTION – B

25. 2

Sol. It will form buffer solution

Let V ml of KCN is added, the total volume = V + 10 ml

$$[HCN] = \frac{20}{V+10}, [KCN] = \frac{5V}{V+10}$$

$$pH = pk_a + \log\frac{[KCN]}{[HCN]}$$

$$9 = 9.3 + \log\frac{5V}{20}$$

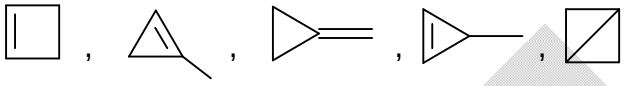
$$\therefore \quad V = 2m\ell$$

26. 5

Sol. PbS, Ag_2S, NiS, CuS, CoS are black coloured solid.

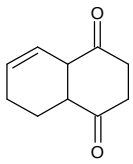
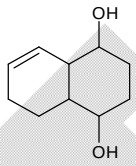
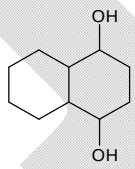
27. 9

Sol. $C-C=C=C$, $C=C-C=C$, $C\equiv C-C-C$, $C-C\equiv C-C$



28. 4

Sol.

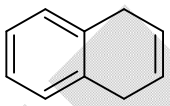
A =  , B =  , C = 

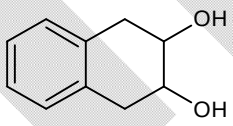
29. 3

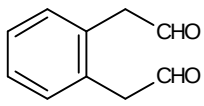
Sol. (vi), (vii) and (viii) have non-superimposable mirror image.

30. 6

Sol.


(A)


(B)


(C)

SECTION – C

31. 1.26

Sol. $T\ell^{3+}_{(aq)} + 2e^- \longrightarrow T\ell^{+}_{(aq)} \quad E^0 = ?$

$T\ell^{+}_{(aq)} + e^- \longrightarrow T\ell_{(s)} \quad \Delta G_1^0 = -F(-0.336) = 0.336F$

$T\ell^{3+}_{(aq)} + 3e^- \longrightarrow T\ell_{(s)} \quad \Delta G_2^0 = -3F(0.728)$

$T\ell^{3+}_{(aq)} + 2e^- \longrightarrow T\ell^{+}_{(aq)} \quad \Delta G^0 = -3FE^0$

$\Delta G^0 = \Delta G_2^0 - \Delta G_1^0$

$-2FE^0 = -3F \times 0.278 - 0.336F$

$= -2.184F - 0.336F \qquad E^0 = \frac{2.52}{2} = 1.26V$

32. 26.96

Sol. From eq.

$I_{2(s)} + 2e^- \longrightarrow 2I^-_{(aq)} \qquad \Delta G^o = -2F(0.54)$

$I_{2(s)} + I^-_{(aq)} \longrightarrow I^-_{3(aq)} \qquad \Delta G^o = -RT\ln(0.459)$

$\therefore$ for $I^-_{3(aq)} + 2e^- \longrightarrow 3I^-_{(aq)} \qquad \Delta G^o = -106.1\,kJ\,mol^{-1}$

The reaction $T\ell^+_{(aq)} + I^-_{3(aq)} + I^-_{(aq)} \rightleftharpoons T\ell I^-_{4(aq)}$ can be formed by following reactions:

$T\ell^+_{(aq)} \longrightarrow T\ell^{3+}_{(aq)} + 2e^- \qquad E^0 = -1.26\,V; \qquad \Delta G^0_1 = 243.1kJ\,mol^{-1}$

$I^-_{3(aq)} + 2e^- \longrightarrow 3I^-_{(aq)} \qquad \Delta G^0_2 = -106.1kJ\,mol^{-1}$

$T\ell^{3+}_{(aq)} + 4I^-_{(aq)} \rightleftharpoons T\ell I^-_{4(aq)} \qquad K_f = 10^{35.7}; \qquad \Delta G^0_3 = -203.8\,kJ\,mol^{-1}$

The net free energy change for the reaction

$T\ell^+_{(aq)} + I^-_{3(aq)} + I^-_{(aq)} \rightleftharpoons T\ell I^-_{4(aq)}$

$\Delta G^0_{Rx} = \Delta G^0_1 + \Delta G^0_2 + \Delta G^0_3$

$= -66.8\,kJmol^{-1}$

$\Delta G^0_{Rx} = -RT\ln K_{eq}$

$\ln K_{eq} = -\dfrac{\Delta G^0_{Rx}}{RT} = \dfrac{66.8\times10^3}{8.314\times298} = 26.96$

33. 92.92

Sol. $2PbS + 3O_2 \longrightarrow 2PbO + 2SO_2$

Moles of PbS = moles of $SO_2 = \dfrac{PV}{RT} = 2.706\times10^{-3}$ (only from two-third of the sample)

So total $n_{PbS} = \dfrac{3}{2}\times2.706\times10^{-3} = 4.06\times10^{-3}$

Mass of PbS $= 4.06\times10^{-3}\times239.2 = 0.971\,g,$

% purity $= \dfrac{W_{PbS}}{W_{gelena}}\times100 = 92.92$

34. 2.90

Sol. Total mass of lead produced = 0.8663 g

Total mole of lead $= \dfrac{0.8663}{207.2}$

$(n_{Pb} + n_{PbS}) = 4.18\times10^{-3} \qquad n_{Pb} = (n_{Pb} + n_{PbS}) - (n_{PbS})$

$= 4.18\times10^{-3} - 4.06\times10^{-3}$

$= 0.12\times10^{-3}$

So % elemental lead $= \dfrac{n_{Pb}}{n_{Pb} + n_{PbS}} = \dfrac{0.12\times10^{-3}}{4.18\times10^{-3}}\times100$

$= 2.8 \approx 2.9\%$

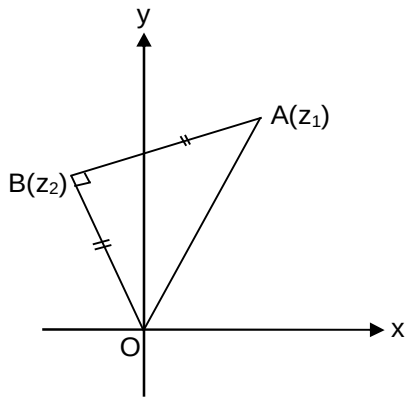
Mathematics

PART – III

SECTION – A

35. A

Sol. $\frac{z_1}{z_2} = z$



$\Rightarrow \quad z^2 - 2z + 2 = 0$

$\Rightarrow \quad z = 1 \pm i$

$\Rightarrow \quad \frac{z_1}{z_2} = 1 \pm i$

$\Rightarrow \quad z_1 = z_2 \pm z_2 i$

$\Rightarrow \quad z_1 - z_2 = \pm z_2 i$

$\Rightarrow \quad z_1 - z_2$ is perpendicular to z_2 and $|z_1 - z_2| = |z_2|$

36. C

Sol. $P_n = {}^{n-2}C_3; Q_n = {}^nC_3 - \left[n + n(n-4)\right]$

or $\quad Q_n = \frac{{}^nC_1 \cdot {}^{n-4}C_2}{3}$

$P_n - Q_n = 6$

$\Rightarrow \quad n = 10$

37. D

Sol. $f'(x) = \frac{1}{x} + \pi\cos(\pi x) + C$

$f'(2) = \frac{1}{2} + \pi + C = \frac{1}{2} + \pi \Rightarrow C = 0$

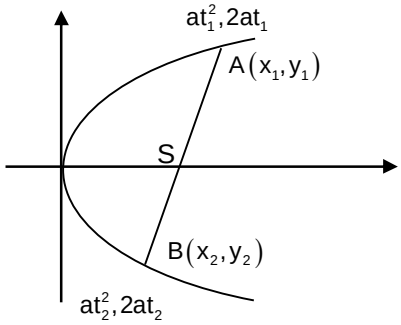
$f(x) = \ln|x| + \sin(\pi x) + C'$

$f(1) = C' = 0$

$f(x) = \ln|x| + \sin(\pi x)]$

38. C

Sol. $y^2 = 4ax, 4a = 2p > 0$



$x_1 = at_1^2, y_1 = 2at_1$

$x_2 = at_1^2, y_2 = 2at_2$

and $t_1t_2 = -1$

$$\text{Ratio} = \frac{4a^2t_1t_2}{a^2t_1^2t_2^2} = -4$$

39. CD

Sol. $P(E_1) = 1 - P(\text{unit's place in both is 1, 2, 3, 4, 6, 7, 8, 9})$

$$P(E_1 = 0 \text{ or } 5) = 1 - \left(\frac{4}{5}\right)^2 = \frac{9}{25}$$

$P(E_2 : 5) = P(1\,3\,5\,7\,9) - P(1\,3\,7\,9)$ for 2 numbers

$$= \frac{1}{4} - \frac{4}{25} = \frac{25-16}{100} = \frac{9}{100}$$

$$\frac{P(E_2)}{P(E_1)} = \frac{9}{100} \cdot \frac{9}{100} \cdot \frac{25}{9} = \frac{1}{4}$$

$P(E_1) = 4P(E_2)$

$\Rightarrow$ (A) is not correct.

$$P\left(\frac{E_2}{E_1}\right) = \frac{P(E_2 \cap E_1)}{P(E_1)} = \frac{P(E_2)}{P(E_1)} = \frac{9}{100} \cdot \frac{25}{9} = \frac{1}{4}$$

$\Rightarrow$(C)

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{P(E_1)}{P(E_2)} = 1$$

$\Rightarrow (D)$

40. ABCD

Sol. Use $R_1 \to R_1 - R_2$ and $R_2 \to R_2 - R_3$ and expand to get $D = 2(1 + \sin 4\theta) = 0$

$$\Rightarrow \quad \theta = \frac{n\pi}{4} - (-1)^n \frac{\pi}{8}$$

$\Rightarrow$ Independent of A

$\Rightarrow$ (A),(B),(C),(D)

41. AC

Sol. $$f(x) = \begin{cases} x^3 & \text{if } x \le 1 \\ x^2 & \text{if } x > 1 \end{cases}; \quad g(x) = \begin{cases} x+2 & -1 < x < 0 \\ x & 0 \le x < 1 \\ x+2 & 2 \le x < 3 \end{cases}$$

$\Rightarrow$ (A),(C)

Also g(x) be continuous at $x \in I$

$g(n) = n^2;\ g(n^+) = n^2;\quad g(n^-) = (n-1)^2 + 1$

For continuity $n^2 = (n-1)^2 + 1$

$\Rightarrow$ $n = 1$

$\Rightarrow$ g(x) is continuous only at one integer x = 1.

SECTION – B

42. 9

Sol. Equation of plane P (with usual $\vec{n}_1$ which n to $\vec{n}$ and $\overline{AB}$

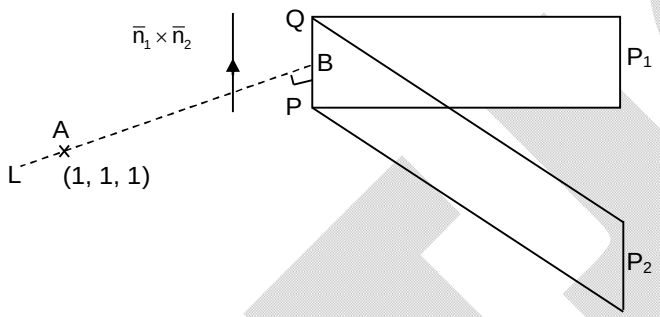
$\overline{n_1} = \overline{n} \times \overline{AB} = (i + j - 3k) \times (i - 3j + k) = 2i + j + k$

$\therefore$ Equation of plane

$P_1 : 2(x-1) + 1(y-2) + 1(z-0) = 0$

$\Rightarrow$ $P_1 : 2x + y + z = 4$

Also, $P_2 : +x + y + z = 5$



Line PQ is parallel to vector

$$\overline{n}_1 \times \overline{n}_2 = \begin{vmatrix} i & j & k \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{vmatrix} = k - j$$

$\therefore$ Equation PQ : $\dfrac{x+1}{0} = \dfrac{y-6}{-1} = \dfrac{z-0}{1} = \lambda$

(Contained P_1 and P_2 in symmetric form)

Let $B(-1, 6-\lambda, \lambda)$ be any point on line PQ

So, $\overrightarrow{AB} = -2\hat{i} + (5-\lambda)\hat{j} + (\lambda - 1)\hat{k}$

Now, $\overrightarrow{AB} \cdot \vec{n}_1 \times \vec{n}_2 = 0$

$\Rightarrow$ $0 - 5 + \lambda + \lambda - 1 = 0$

$\Rightarrow$ $\lambda = 3$

$\therefore$ $B \equiv (-1, 3, 3)$

So, equation of line L is $\dfrac{x+1}{1+1} = \dfrac{y-3}{1-3} = \dfrac{z-3}{1-3}$

i.e., $\dfrac{x+1}{-1} = \dfrac{y-3}{1} = \dfrac{z-3}{1}$

Hence $\left(\dfrac{2y_1 + z_1}{bc} + \dfrac{6+3}{1}\right) = 9$

43. 5

Sol. $J=\int_0^1\left(1-x^{50}\right)^{100}x^{100}dx$

$$=\left((1-x^{50})^{100}\frac{x^{101}}{101}\right)_0^1-\int_0^1 100\left(1-x^{50}\right)^{99}\left(-50x^{49}\right)\frac{x^{101}}{101}dx$$

$$=0+\frac{5000}{101}\int_0^1\left(1-x^{50}\right)^{99}\left(1-\left(1-x^{50}\right)\right)x^{100}dx$$

$$=\frac{5000}{101}\left(\int_0^1\left(1-x^{50}\right)^{99}x^{100}dx-\int_0^1\left(1-x^{50}\right)^{100}x^{100}dx\right)$$

$$J+\frac{5000}{101}J=\frac{500}{101}I$$

$$\frac{5101}{101}J=\frac{5000}{101}I$$

$\Rightarrow \quad \frac{I}{J}=\frac{5101}{5000}=\frac{m}{n}$

$\therefore \quad \frac{m-1-n}{20}=\frac{5101-1-5000}{20}=5$

44. 4

Sol. $f(x)=\ln\left(1+x^2\right)+\tan^{-1}x$

$$g(f(x))=x$$

$$g'(f(x))f'(x)=1$$

$\Rightarrow \quad g'(f(x))=\frac{1}{f'(x)} \quad \Rightarrow \quad g''(f(x))f'(x)=-\frac{f''(x)}{(f'(x))^2}$

$\Rightarrow \quad g''(f(x))=-\frac{f''(x)}{(f'(x))^3}$

$$f(x)=\ln\left(2e^{\pi/4}\right)$$

$$=\ln\left(1+x^2\right)\tan^{-1}x$$

$\Rightarrow \quad \ln 2+\frac{\pi}{4}=\ln\left(1+x^2\right)+\tan^{-1}x$

$\Rightarrow \quad x=1$

$$g''(\ln(2e^{\pi/4}))=\frac{-f''(1)}{(f'(1))^3}$$

$$f'(x)=\frac{1}{1+x^2}+\frac{2x}{1+x^2}+\frac{2x+1}{1+x^2}$$

$\Rightarrow \quad f'(1)=\frac{3}{2}$

$$f''(x)=\frac{\left(1-x^2\right)2-(2x+1)\cdot 2x}{\left(1-x^2\right)^2}$$

$$=\frac{2\left(1-x-x^2\right)}{\left(1+x^2\right)^2}$$

$\Rightarrow \quad f''(1) = \dfrac{-1}{2}$

$\therefore \quad g''(\ln 2e^{\pi/4}) = \dfrac{2}{27} = \dfrac{-4}{27}$

$\Rightarrow \quad \left|27 \cdot g''(\ln(2e^{\pi/4}))\right| = 4$

45. 32

Sol. We have, $a_n = \tan^{-1}\left(\dfrac{3}{9n^2+3n-1}\right)$

$= \tan^{-1}\left(\dfrac{3}{1+(3n+2)(3n-1)}\right)$

$= \tan^{-1}\left(\dfrac{(3n+2)(3n-1)}{1+(3n+2)(3n-1)}\right)$

$= \tan^{-1}(3n+2) - \tan^{-1}(3n-1)$

$\therefore$ Sum of first 10 terms

$= \sum_{r=1}^{10} a_r = \sum_{r=1}^{10}\left(\tan^{-1}(3r+2) - \tan^{-1}(3r-1)\right)$

$= \left(\tan^{-1}5 - \tan^{-1}2\right) + t\left(\tan^{-1}8 - \tan^{-1}5\right) + ... + \left(\tan^{-1}32 - \tan^{-1}29\right)$

$= \tan^{-1}32 - \tan^{-1}2 = \tan^{-1}\left(\dfrac{32-2}{1+32.2}\right)$

$= \tan^{-1}\dfrac{30}{65} = \tan^{-1}\left(\dfrac{6}{13}\right) = \cot^{-1}\left(\dfrac{13}{6}\right) = \cot^{-1}\left(\dfrac{m}{n}\right)$

$\therefore \quad m = 13$ and $n = 6$

Hence, $(2m+n) = 32$

46. 816

Sol. $f(x) = a_0 + a_1(1+x) + a_2(1+x)^2 + ...$

$f'(x) = a_1 + 2a_2(1+x) + 3a_3(1+x)^2 + ...$

$f''(x) = 2a_2 + 6a_3(1+x) + ...$

$f''(-1) = 2a_2$ (Note)

Now, $f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17}$

$f'(x) = -1 + 2x - 3x^2 + ... + 16x^{15} - 17x^{16}$

$f''(x)]_{x=-1} = 1\cdot 2 + 2\cdot 3 + 3\cdot 4 + ... + 16\cdot 17$

$= \sum_{r=1}^{16} r(r+1) = \sum r^2 + \sum r$

$= \dfrac{16.17.33}{6} + \dfrac{16.17}{2} = \dfrac{16.17}{2}[11+1]$

$f''(-1) = 16.17.6$

$2a_2 = 16.17.6$

$a_2 = 16.51 = 816$

47. 10

Sol. Suppose roots be $a^2,(a+1)^2$, where $\alpha \in N$

So, the equation is $\left(x-a^2\right)\left[x-(a+1)^2\right]=0$

Given, $\sqrt{a^2(a+1)^2}=1+\left|(a+1)^2-a^2\right|$

$\Rightarrow \quad a(a+1)=1+(2a+1) \qquad (\text{As } a \in N)$

$\Rightarrow \quad a^2+a=1+2a+1$

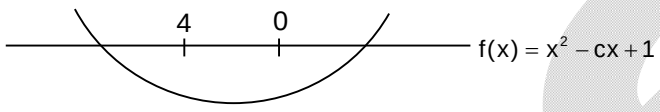
$\Rightarrow \quad a^2-a-2=0$

$\Rightarrow \quad (a-2)(a+1)=0$

$\Rightarrow \quad a=2 \text{ or } -1$

(Reject, $a=-1$ as $a \in N$)

Hence, roots are 4 and 9.



Let $f(x)=x^2-cx+1$

So, $f(4)<0 \Rightarrow 17---4c<0 \Rightarrow c>\frac{17}{4}$ ...(i)

and $f(9)<0 \Rightarrow 82-9c<0 \Rightarrow c>\frac{82}{9}$ ...(ii)

$\therefore$ Eqns. (i) $\cap$ (ii) gives $c>\frac{82}{9}$

$\Rightarrow$ Hence, minimum integral value of c = 10.

SECTION – C

48. 17.00

Sol. $\int \frac{f(x)}{x^3-1}dx=\ln\left|\frac{x^2+x+1}{x-1}\right|+\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$

Differentiating both sides, we get

$$\frac{f(x)}{x^3-1}=\frac{x-1}{x^2+x+1}\cdot\frac{(x-1)(2x+1)-\left(x^2+x+1\right)\cdot 1}{(x-1)^2}+\frac{2}{\sqrt{3}}\cdot\frac{1}{1+\left(\frac{2x+1}{\sqrt{3}}\right)^2}\cdot\frac{2}{\sqrt{3}}$$

$$=\frac{2x^2-x-1-x^2-x-1}{x^3-1}+\frac{4}{3}\cdot\frac{3}{3+\left(4x^2+4x+1\right)}$$

$$=\frac{x^2-2x-2}{x^3-1}+\frac{1}{x^2+x+1}$$

$f(x)=\left(x^2-2x-2\right)+(x-1)=x^2-x-3$

$\Rightarrow f(5)=5^5-5-3=17$

49. 3.00

Sol. $I=\int\frac{1-6\operatorname{cosec}x}{6+f(\sin x)}d(\sin x)$

$$= \int \frac{1-\frac{6}{\sin x}}{6+\sin^2 x-\sin x-3} d(\sin x)$$

$$= \int \frac{1-\frac{6}{\sin x}}{\sin^2 x-\sin x+3} d(\sin x) \qquad (\text{Put } \sin x = t)$$

$$= \int \frac{-\frac{6}{t}}{t^2-t+3} dt = \int \frac{\frac{1}{t^2}-\frac{6}{t^3}}{1-\frac{1}{t}+\frac{3}{t^2}} dt$$

$$= \ln\left(1-\frac{1}{t}+\frac{3}{t^2}\right)+K = \ln\left(1-\frac{1}{\sin x}+\frac{3}{\sin^2 x}\right)+K$$

$$\Rightarrow g(x) = \ln\left(1-\frac{1}{\sin x}+\frac{3}{\sin^2 x}\right)$$

$$\therefore g(x) = \ln\left(1-\frac{1}{\sin t}+\frac{3}{\sin^2 t}\right)$$

Now, $\lim\limits_{t\to\pi/2} g(t) = \ln 3$

50. 4.00

Sol. $$f(t) = \sum_{k=0}^{\infty} \frac{1}{(t+2)^k} = \frac{1}{1-\frac{1}{(t+2)}} = \frac{t+2}{t+1}$$

$$f(0) = 2$$

$$S = 1+\frac{2}{2}+\frac{3}{2^2}+\frac{4}{2^3}+\frac{5}{2^4}+.....$$

$$\frac{S}{2} = \frac{1}{2}+\frac{2}{2^2}+\frac{3}{2^3}+.....$$

$$\frac{S}{2} = 1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+.....$$

$$\Rightarrow \frac{S}{2} = \frac{1}{1-\frac{1}{2}}$$

$$\Rightarrow S = 4$$

51. 1.00

Sol. $$f(k) = \frac{k+2}{(k+1)}$$

$$\frac{f(k)}{k(k+2)} = \frac{1}{k(k+1)}$$

$$\sum_{k=1}^{\infty} \frac{f(k)}{k(k+1)} = \sum_{k=1}^{\infty}\left(\frac{1}{k}=\frac{1}{k+1}\right) = 1$$